

Roadmap to Learn C++

Phase 1: Basics of C++

Topic	What You'll Learn
✓ 1. Hello World	Basic structure of a C++ program
✓ 2. Variables & Data Types	int, float, char, string, bool
✓ 3. Input & Output	cin, cout
✓ 4. Operators	Arithmetic, Comparison, Logical
✓ 5. Control Flow	if, else, switch, loops (for, while, do-while)
✓ 6. Functions	Declaring, calling, parameters, return values
✓ 7. Arrays & Strings	Single-dimension, multi-dimension, C-strings vs std::string

Phase 2: Object-Oriented Programming (OOP)

Topic	What You'll Learn
✓ 1. Classes & Objects	Blueprint and instances
✓ 2. Constructors	Default, parameterized
✓ 3. Inheritance	public, protected, private inheritance
✓ 4. Polymorphism	Function overloading, overriding, virtual functions
✓ 5. Encapsulation	Access modifiers (public, private, protected)
✓ 6. Abstraction	Using interfaces and abstract classes

Phase 3: Advanced C++

Topic	What You'll Learn
 1. Pointers	Basics, arrays and pointers, pointer arithmetic
 2. Dynamic Memory	new, delete, heap allocation
 3. File Handling	Read/write files using fstream
 4. STL (Standard Template Library) vector, map, set, stack, queue	